

Negative Binomial Distribution:

An experiment in which the trials are repeated until a fixed number of successes occurs is called negative binomial experiment.

Let x be the number of failures preceding the k^{th} success in $x+k$ trials. The random variable x which represents the number of trials needed to produce k successes is called a negative Binomial random variable.

In BD, the number of trials n is kept fixed and the number of successes x is random variable but in NBD, the number of trials is a random variable and the number of successes is fixed.

Definition:

A random variable (r.v) x is said to follow a negative Binomial distribution with parameter k and p if its probability mass function (pmf) is defined as;

$$P(X=x) = \binom{x+k-1}{k-1} p^k q^x, \quad x=0,1,2,3,\dots \text{ and } k > 0$$

$$0, \text{ otherwise}$$

where,

x = no. of failures

k = no. of successes (no. of target)

n = total no. of trials

p = prob. of success

q = prob. of failure

$$BD = P(X=x) = {}^nC_x p^x q^{n-x}$$

$${}^nC_x = \binom{n}{x}$$

$$P(X=x) = {}^n C_x p^x q^{n-x}$$

S S F S S F S S F $n=9$

P P 9 P P 9 P P 9

$${}^n C_x \cdot p^6 q^3 = p^x q^{9-x} = p^x q^{n-x}$$

For binomial distⁿ

$$n \leq 20$$

$$p \rightarrow 1/2$$

$$q \rightarrow 1/2$$

Derivation of Binomial Distribution:

Consider a Binomial experiment consists of n independent trials and the probability of success is a constant for each trial.

Let x be the number of failures preceding the k th success in $x+k$ trials. Here, $n = x+k$ trials are required to produce k number of successes. The probability that for $x+k-1$ number of trials and $k-1$ number of successes given as $\binom{x+k-1}{k-1} p^{k-1} q^x$ and the

last trial must be success with prob. p then the prob. of x failures preceding the k th success in $x+k$ trials is given by;

$$P(X=x) = \binom{x+k-1}{k-1} p^{k-1} q^x \cdot p$$

$$\boxed{P(X=x) = \binom{x+k-1}{k-1} p^k q^x}$$

Moments of Negative Binomial distⁿ:

Let $X \sim NB(k, p)$ then its pmf is given as;

$$P(X=x) = \binom{x+k-1}{k-1} p^k q^x$$

The r th moment about origin of X is given by;

$$\mu_r' = E(X^r) = \sum_{x=0}^{\infty} x^r P(X=x)$$

$$\mu_1' = \sum_{x=0}^{\infty} x \binom{x+k-1}{k-1} p^k q^x$$

If $r=1$,

$$\mu_1' = \sum_{x=0}^{\infty} x \binom{x+k-1}{k-1} p^k q^x$$

$$\mu_1' = E(x) = \sum_{x=0}^{\infty} x \frac{(x+k-1)!}{(k-1)!(x+k-1-k+1)!} p^k q^x$$

$$= \sum_{x=0}^{\infty} x \frac{(x+k-1)!}{x!(k-1)!} p^k q^x$$

$$= \sum_{x=1}^{\infty} x \frac{(x+k-1)!}{x(x-1)!(k-1)!} p^k q^x$$

$$= \sum_{x=1}^{\infty} \frac{qk(x+k-1)!}{p(x-1)!k(k-1)!} p^{k+1} q^{x-1}$$

$$= \frac{qk}{p} \left[\sum_{x=1}^{\infty} \frac{(x+k-1)!}{(x-1)!k(k-1)!} p^{k+1} q^{x-1} \right]$$

$$= \frac{kq}{p} \left[\sum_{x=1}^{\infty} \binom{x+k-1}{k} p^{k+1} q^{x-1} \right]$$

$$\boxed{E(x) = \frac{kq}{p}}$$

Again,

$$\mu_2' = E(x^2) = \sum_{x=0}^{\infty} x^2 p(x=x)$$

$$= \sum_{x=0}^{\infty} \{x(x-1) + x\} \binom{x+k-1}{k-1} p^k q^x$$

$$E(x^2) = \sum_{x=0}^{\infty} x(x-1) \binom{x+k-1}{k-1} p^k q^x + \left[\sum_{x=0}^{\infty} x \binom{x+k-1}{k-1} p^k q^x \right]$$

$$E(x^2) = \sum_{x=0}^{\infty} x(x-1) \frac{(x+k-1)!}{(k-1)!(x+k-1-k+1)!} p^k q^x + \frac{kq}{p}$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{(x+k-1)!}{(k-1)!x!} p^k q^x + \frac{kq}{p}$$

$$= \sum_{x=2}^{\infty} x(x-1) \frac{(x+k-1)!}{x(x-1)(x-2)!(k-1)!} p^k q^x + \frac{kq}{p}$$

$$= \sum_{x=2}^{\infty} \frac{(x+k-1)!}{(x-2)!(k-1)!} p^k q^x + \frac{kq}{p}$$

$$= \sum_{x=2}^{\infty} \frac{q^2 k(k+1) (x+k-1)!}{p^2 k(k+1)(k-1)!(x-2)!} p^{k+2} q^{x-2} + \frac{kq}{p}$$

$$= \frac{k(k+1)q^2}{p^2} \left[\sum_{x=2}^{\infty} \frac{(x+k-1)!}{(k-2)!(x-2)!} p^{k+2} q^{x-2} \right] + \frac{kq}{p}$$

$$= \frac{k(k+1)q^2}{p^2} + \frac{kq}{p}$$

$$\therefore V(x) = E(x^2) - [E(x)]^2$$

$$= \frac{k(k+1)q^2}{p^2} + \frac{kq}{p} - \left(\frac{kq}{p} \right)^2$$

$$V(x) = \frac{k(k+1)q^2}{p^2} + \frac{kq}{p} - \frac{k^2 q^2}{p^2}$$

$$\boxed{V(x) = \frac{kq}{p^2}} \quad \because p+q=1$$

As $k > 1$, $p < 1$, $q < 1$

$$E(X) < V(X)$$

In NBD, mean < variance

Poisson Distribution as a limiting case of NBD:

Negative Binomial distribution tends to Poisson distribution as $p \rightarrow 0$, and $k \rightarrow \infty$ such that $kp = \lambda$.

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x=0, 1, 2, \dots, \infty$$

$$p \rightarrow 0$$

$$n = 1000 \rightarrow \infty$$

$$q \rightarrow 1$$

$$x = 2$$

$$p+q=1$$

$$p = 2/1000 = 0.002 \rightarrow 0$$

$$q = 998/1000 = 0.998 \rightarrow 1$$

Moment generating function of NBD:

Let $X \sim NB(k, p)$ then its pmf is given as,

$$P(X=x) = \binom{x+k-1}{k-1} p^k q^x, \quad x=0, 1, 2, \dots, (k > 0)$$

If $p = \frac{1}{Q}$ and $q = \frac{P}{Q}$ so that $q+p=1$
 $Q = 1+p$

$$P(X=x) = \binom{x+k-1}{k-1} \left(\frac{1}{Q}\right)^k \left(\frac{P}{Q}\right)^x$$

$$P(X=x) = \binom{x+k-1}{k-1} Q^{-k} \left(\frac{P}{Q}\right)^x$$

The moment generating function of $X \sim NB(k, p)$ is;

$$M_x(t) = E(e^{tx})$$

$$= \sum_{x=0}^{\infty} e^{tx} p(x=x)$$

$$= \sum_{x=0}^{\infty} e^{tx} \binom{x+k-1}{k-1} \phi^{-k} \left(\frac{p}{\phi}\right)^x$$

$$E(x) = \sum x p(x=x)$$

$$M_x(t) = \phi^{-k} \sum_{x=0}^{\infty} \binom{x+k-1}{k-1} \left(\frac{e^t p}{\phi}\right)^x$$

$$M_x(t) = \phi^{-k} \left(1 - \frac{pe^t}{\phi}\right)^{-k}$$

$$\therefore \sum_{x=0}^{\infty} \binom{-k}{x} (-q)^x = (1-q)^{-k}$$

$$M_x(t) = (\phi - pe^t)^{-k}$$

Now, the moment about origin of NBD can be obtained by;

$$\mu_1' = \frac{d^r M_x(t)}{dt^r} \bigg|_{t=0}$$

$$\text{Put } r=1,$$

$$\mu_1' = E(x) = \frac{d M_x(t)}{dt} \bigg|_{t=0}$$

$$= \frac{d (\phi - pe^t)^{-k}}{dt} \bigg|_{t=0}$$

$$= -k (\phi - pe^t)^{-k-1} (-pe^t) \bigg|_{t=0}$$

$$= k pe^t (\phi - pe^t)^{-k-1} \bigg|_{t=0}$$

$$\mu_1' = kp$$

Since $p = \frac{1}{q} \Rightarrow q = \frac{1}{p}$

$$q = \frac{p}{q}$$

$$p = q q$$

$$\boxed{p = \frac{q}{p}}$$

$$\mu_1' = k \cdot \frac{q}{p}$$

$$\text{Mean} = \boxed{\mu_1' = \frac{kq}{p}}$$

Similarly,

$$\mu_2' = \frac{d^2 M_x(t)}{dt^2} \bigg|_{t=0}$$

$$= \frac{d}{dt} \left[k p e^t (q - p e^t)^{-k-1} \right] \bigg|_{t=0}$$

$$= k p e^t (q - p e^t)^{-k-1} + k p e^t (-k-1) (q - p e^t)^{-k-2} (-p e^t) \bigg|_{t=0}$$

$$\boxed{\mu_2' = k p + k(k+1)p^2}$$

$$\text{Variance} = \mu_2 = \mu_2' - (\mu_1')^2$$

$$= k p + k(k+1)p^2 - (k p)^2$$

$$= k p + k p^2$$

$$= k p (1 + p)$$

$$= k p q$$

$$\therefore V(x) = k p q = \frac{k q}{p^2}$$

Similarly,

$$\beta_1 (\text{skewness}) = \frac{\mu_3^2}{\mu_2^3} = \frac{(1+q)^2}{kq}$$

$$\gamma_1 = \sqrt{\beta_1} = \frac{1+q}{\sqrt{kq}}$$

$$\beta_2 (\text{Kurtosis}) = \frac{\mu_4}{\mu_2^2} = \frac{p^2 + 3q(k+2)}{kq}$$

$$\gamma_2 = \beta_2 - 3 = \frac{p^2 + 6}{kq}$$

→ Right skewed and leptokurtic

Negative Binomial distribution tends to Poisson distⁿ as $p \rightarrow 0$ and $k \rightarrow \infty$ such that $kp = \lambda$.

Proof:

Let $X \sim NB(k, p)$ then its pmf is;

$$P(X=x) = \binom{x+k-1}{k-1} p^k q^x$$

If $p = \frac{\lambda}{q}$ and $q = \frac{p}{\lambda}$ then

$$P(X=x) = \binom{x+k-1}{k-1} \left(\frac{\lambda}{q}\right)^k \left(\frac{p}{\lambda}\right)^x$$

$$= \lim_{\substack{k \rightarrow \infty \\ p \rightarrow 0}} \binom{x+k-1}{k-1} q^{-k} \left(\frac{p}{\lambda}\right)^x$$

$$= \lim_{\substack{k \rightarrow \infty \\ p \rightarrow 0}} \left(\frac{x+k-1}{x} \right) q^{-k} \left(\frac{p}{q} \right)^x$$

$$= \lim_{k \rightarrow \infty} \frac{(x+k-1)(x+k-2) \dots (k+1)k}{x!} (1+p)^{-k} \left(\frac{p}{1+p} \right)^x$$

$$= \lim_{k \rightarrow \infty} \left[\frac{1}{x!} \left(1 + \frac{x-1}{k} \right) \left(1 + \frac{x-2}{k} \right) \dots \left(1 + \frac{1}{k} \right) k^x (1+p)^{-k} \left(\frac{p}{1+p} \right)^x \right]$$

$$= \frac{1}{x!} \lim_{k \rightarrow \infty} \left[(1+p)^{-k} \left(\frac{kp}{1+p} \right)^x \right]$$

$$\therefore kp = \lambda$$

$$p = \lambda/k$$

$$= \frac{\lambda^x}{x!} \lim_{k \rightarrow \infty} \left(1 + \frac{\lambda}{k} \right)^{-k} \lim_{k \rightarrow \infty} \left(1 + \frac{\lambda}{k} \right)^{-x}$$

$$= \frac{\lambda^x}{x!} e^{-\lambda} \cdot 1$$

$$\boxed{P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}}$$

Thus is the proof of Poisson distⁿ.

Similarly,

$$\beta_1 \text{ (skewness)} = \frac{\mu_3^2}{\mu_2^3} = \frac{(1+q)^2}{kq}$$

$$\gamma_1 = \sqrt{\beta_1} = \frac{1+q}{\sqrt{kq}}$$

$$\beta_2 \text{ (Kurtosis)} = \frac{\mu_4}{\mu_2^2} = \frac{p^2 + 3q(k+2)}{kq}$$

$$\gamma_2 = \beta_2 - 3 = \frac{p^2 + 6}{kq}$$

↳ Right skewed and leptokurtic

Negative Binomial distribution tends to Poisson distⁿ as $p \rightarrow 0$ and $k \rightarrow \infty$ such that $kp = \lambda$.

Proof:

Let $X \sim NB(k, p)$ then its pmf is;

$$P(X=x) = \binom{x+k-1}{k-1} p^k q^x$$

If $p = \frac{1}{q}$ and $q = \frac{p}{q}$ then

$$P(X=x) = \binom{x+k-1}{k-1} \left(\frac{1}{q}\right)^k \left(\frac{p}{q}\right)^x$$

$$= \lim_{\substack{k \rightarrow \infty \\ p \rightarrow 0}} \binom{x+k-1}{k-1} q^{-k} \left(\frac{p}{q}\right)^x$$

$$= \lim_{\substack{k \rightarrow \infty \\ p \rightarrow 0}} \binom{x+k-1}{x} q^{-k} \left(\frac{p}{q}\right)^x$$

$$= \lim_{k \rightarrow \infty} \frac{(x+k-1)(x+k-2) \dots (k+1)k}{x!} (1+p)^{-k} \left(\frac{p}{1+p}\right)^x$$

$$= \lim_{k \rightarrow \infty} \left[\frac{1}{x!} \left(1 + \frac{x-1}{k}\right) \left(1 + \frac{x-2}{k}\right) \dots \left(1 + \frac{1}{k}\right) k^x (1+p)^{-k} \left(\frac{p}{1+p}\right)^x \right]$$

$$= \frac{1}{x!} \lim_{k \rightarrow \infty} \left[(1+p)^{-k} \left(\frac{kp}{1+p}\right)^x \right]$$

$$\therefore kp = \lambda$$

$$p = \lambda/k$$

$$= \frac{\lambda^x}{x!} \lim_{k \rightarrow \infty} \left(1 + \frac{\lambda}{k}\right)^{-k} \lim_{k \rightarrow \infty} \left(1 + \frac{\lambda}{k}\right)^{-x}$$

$$= \frac{\lambda^x}{x!} e^{-\lambda} \cdot 1$$

$$\boxed{P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}}$$

Thus is the proof of Poisson distⁿ.

Recurrence Relation for NBD:

Let $X \sim NB(k, p)$ then its pmf is

$$P(X=x) = \binom{x+k-1}{k-1} p^k q^x$$

$$\text{and } P(X=x+1) = \binom{x+k}{k-1} p^k q^{x+1}$$

$$\text{Now, } \frac{P(X=x+1)}{P(X=x)} = \frac{\binom{x+k}{k-1} p^k q^{x+1}}{\binom{x+k-1}{k-1} p^k q^x}$$

$$\frac{P(X=x+1)}{P(X=x)} = \frac{\binom{x+k}{k-1} q}{\binom{x+k-1}{k-1}}$$

$$= \frac{(x+k)!}{(k-1)! (x+k-k+1)!} q$$

$$= \frac{(x+k)!}{(x+k-1)!} q$$

$$= \frac{(x+k)!}{(x+k-1)!} q$$

$$= \frac{(x+k)!}{(x+k-1)!} q$$

$$= \frac{(x+k)!}{(x+k-1)!} q$$

$$= \frac{(x+k)!}{(x+k-1)!} q$$

$$= \frac{(x+k)!}{(x+k-1)!} q$$

$$= \frac{(x+k)(x+k-1)!}{(x+k-1)!} q$$

$$= (x+k) q$$

$$\boxed{\frac{P(X=x+1)}{P(X=x)} = \frac{(x+k) q}{(x+k-1)!}}$$

This is the recurrence relation for probabilities of NBD for $x = 0, 1, 2, \dots$ and $k > 0$. This recurrence relation formula is useful for fitting the NBD to the given empirical data.

Properties of NBD

(i) In NBD, the number of trials is random variable and number of successes is fixed.

$$(ii) E(x) = \frac{kq}{p}, \quad v(x) = \frac{kq}{p^2}$$

In NBD, mean is less than variance $E(x) < v(x)$.

(iii) MGF of NBD,

$$M_x(t) = (q - pe^t)^{-k}$$

where, $q = \frac{1}{p}$ and $q = \frac{p}{q}$ so that $q - p = 1$

(iv) Let $X \sim NBD(k, p)$ then; moment coefficient of skewness $(\gamma_1) = \frac{1+q}{\sqrt{kq}}$

$$\text{Moment coefficient of kurtosis } (\gamma_2) = \frac{p^2 + 6q}{kq}$$

So the NBD is positively skewed and leptokurtic.

(v) The NBD follows additive property for independent random variable.

(vi) The recurrence relation for NBD is

$$p(x = x+1) = \left(\frac{x+k}{x+1} \right) p(x=x) q$$

(vii) In NBD, NBD tends to Poisson distribution as $p \rightarrow 0$, $k \rightarrow \infty$ and $kp = \lambda$ (finite)

Example:

The probability of hitting target at any trial is 0.2. If a shooter aims at a target, find the probability that the fifth fire is second hit.

Solⁿ:

Given,

$$n = \text{Total number of fire (trials)} = 5$$

$$k = \text{no. of hits (success)} = 2$$

$$p = \text{prob. of hitting the target at any trial} = 0.2$$

$$q = 0.8$$

$$x + k = 5$$

$$x = 5 - k$$

$$x = 3$$

The prob. of x misses preceding the second hit is given by;

$$P(X=x) = \binom{x+k-1}{k-1} p^k q^x$$

$$P(X=3) = \binom{3+2-1}{2-1} (0.2)^2 (0.8)^3$$

$$P(X=3) = 0.08192$$

Thus, the prob. that fifth fire is second hit is the prob of 3 misses preceding the second hit.

Geometric Distribution:

The random variable x , the number of failures before the first success is the geometric variable which follows a distribution is called geometric distribution.

Definition:

A random variable x is said to have geometric distribution with parameter p , if it assumes only non-negative values and its prob. mass function (pmf) is given as;

$$P(x=x) = q^x p \quad , \quad x=0,1,2,\dots$$

$$0 \quad ; \text{ otherwise}$$

Moments of Geometric Distribution:

The r th moment about origin of geometric distribution with parameter p is given by;

$$\mu_r' = E(x^r) = \sum_{x=0}^{\infty} x^r p(x=x) = \sum_{x=0}^{\infty} x^r q^x p$$

Put $r=1$,

$$\mu_1' = E(x^1) = \sum_{x=0}^{\infty} x q^x p = p \sum_{x=0}^{\infty} x q^x$$

$$= p q \sum_{x=0}^{\infty} x q^{x-1}$$

$$\mu_1' = p q [1 + 2q + 3q^2 + 4q^3 + \dots]$$

$$\text{Let } S = 1 + 2q + 3q^2 + 4q^3 + \dots$$

$$Sq = \quad \quad q + 2q^2 + 3q^3 + \dots$$

$$S(1-q) = 1 + q + q^2 + q^3 + \dots$$

$$S(1-q) = (1-q)^{-1}$$

$$\Rightarrow S(1-q) = \frac{1}{(1-q)}$$

$$\Rightarrow S = \frac{1}{(1-q)^2} = \frac{1}{p^2}$$

$$\mu_1' = \frac{pq}{p^2}$$

$$\mu_1' = \frac{q}{p}$$

$$\boxed{E(x) = \frac{q}{p}}$$

Again,

$$\begin{aligned} \mu_2' = E(x^2) &= E[x(x-1) + x] \\ &= E[x(x-1)] + E(x) \end{aligned}$$

$$E(x^2) = E[x(x-1)] + \frac{q}{p} \quad \text{--- (A)}$$

$$\text{Here, } E[x(x-1)] = \sum_{x=0}^{\infty} x(x-1) \cdot p(x=r)$$

$$= \sum_{x=0}^{\infty} x(x-1) p q^x$$

$$= 2pq^2 \sum_{x=2}^{\infty} \frac{x(x-1)}{2} q^{x-2}$$

$$= 2pq^2 [1 + 3q + 6q^2 + \dots]$$

$$= 2pq^2 \frac{1}{(1-q)^3}$$

$$= 2p^2 \cdot 2pq^2 \times \frac{1}{p^3}$$

$$= \frac{2q^2}{p^2}$$

$$\boxed{E[x(x-1)] = \frac{2q^2}{p^2}}$$

Put in (A),

$$\mu_2' = E(x^2) = \frac{2q^2}{p^2} + \frac{q}{p}$$

$$\begin{aligned} \text{Now, variance } (\mu_2) &= \mu_2' - (\mu_1')^2 \\ &= \frac{2q^2}{p^2} + \frac{q}{p} - \left(\frac{q}{p}\right)^2 \end{aligned}$$

$$\therefore V(x) = \frac{q}{p^2} \quad E(x) < V(x)$$

Moment Generating Function of G.D:

Let $X \sim G(p)$ then its pmf is;

$$p(X=x) = q^x p$$

and moment generating function of X is given by;

$$M_X(t) = E(e^{tx}) = \sum_{x=0}^{\infty} e^{tx} p(x=x)$$

$$= \sum_{x=0}^{\infty} e^{tx} q^x p$$

$$= p \sum_{x=0}^{\infty} (qe^t)^x$$

$$= p \frac{1}{(1-qe^t)} = p(1-qe^t)^{-1}$$

$$\therefore M_X(t) = p(1-qe^t)^{-1}$$

$$\text{Now, } \mu_4' = \frac{d}{dt} M_X(t) \Big|_{t=0}$$

$$= \frac{d}{dt} p(1-qe^t)^{-1} \Big|_{t=0}$$

$$\mu_1' = (-1)p(-qe^t)(1-qe^t)^{-2} \Big|_{t=0}$$

$$= pqe^t(1-qe^t)^{-2} \Big|_{t=0}$$

$$= pq(1-q)^{-2}$$

$$= \frac{pq}{p^2}$$

$$\boxed{E(x) = \mu_1' = \frac{q}{p}}$$

$$\text{For } \mu_2' = \frac{d^2}{dt^2} m_x(t) \Big|_{t=0}$$

$$= \frac{d}{dt} [pqe^t(1-qe^t)^{-2}] \Big|_{t=0}$$

$$= pq [e^t(-2)(1-qe^t)^{-3}(-qe^t) + (1-qe^t)^{-2}e^t] \Big|_{t=0}$$

$$= pq [2q(1-q)^{-3} + (1-q)^{-2}]$$

$$= pq \left[\frac{2q}{p^3} + \frac{1}{p^2} \right]$$

$$\mu_2' = \frac{2pq^2}{p^3} + \frac{pq}{p^2}$$

$$V(x) = \mu_2 = \mu_2' - (\mu_1')^2 = \frac{2pq^2}{p^3} + \frac{pq}{p^2} - \left(\frac{q}{p}\right)^2$$

$$\therefore \boxed{V(x) = \mu_2 = \frac{q}{p^2}}$$

Mode:

Mode is the value of the variable x for which the geometric probability function $p(x=x) = pq^x$ is maximum. So the mode is the solution of

$$\frac{d}{dp} p(x) = 0 \quad \text{and} \quad \frac{d^2 p(x)}{dp^2} < 0$$

Thus,

$$\frac{d q^x p}{dp} = 0$$

$$\Rightarrow \frac{d (1-p)^x p}{dp} = 0$$

$$\Rightarrow (1-p)^x + p x (1-p)^{x-1} (-1) = 0$$

$$\text{on } (1-p)^{x-1} [(1-p) - p \cdot x] = 0$$

$$\text{on } (1-p-xp) = 0$$

$$\therefore \boxed{p = \frac{1}{x+1}}$$

$$\text{and } \frac{d^2 p q^x}{dp^2} = \frac{d}{dp} \left[(1-p)^x - x p (1-p)^{x-1} \right]$$

$$= x (1-p)^{x-1} - (-1) - p x (x-1) (1-p)^{x-2} (-1)$$

$$= -2x (1-p)^{x-1} + x p (x+1) (1-p)^{x-2}$$

Hence, mode geometric distance

$$(m_0) = \frac{1}{(1+x)}$$

Ans

Lack of memory property of Geometric distribution

Let x denotes the waiting time for first occurrence of an event A . Suppose the event A can occur at one of time $t = 0, 1, 2, \dots, \infty$. Then, x follows a geometric distribution with pmf

$$p(x=t) = q^t p, \quad t = 0, 1, 2, \dots, \infty$$

Again,

the event A does not occur before k times i.e. $x \geq k$. Then the additional time needed for A to occur. Then the ~~additional time to wait~~ has same distribution as initial time to wait.

Let $y = x - k$ be the amount of additional time needed for A to occur. Then the additional time to wait has same distribution as initial time to wait.

$$\text{i.e., } P(y=t | x \geq k) = P(x=t) = q^t p.$$

Proof:

We have,

$$P(x \geq k) = \sum_{x=k}^{\infty} P(x=x) = \sum_{x=k}^{\infty} q^x p$$

$$= p \sum_{x=k}^{\infty} q^x$$

$$= p (q^k + q^{k+1} + \dots)$$

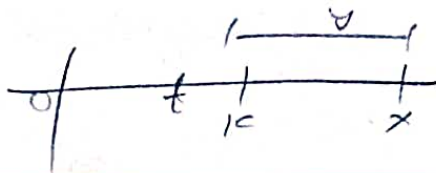
$$= p q^k (1 + q + q^2 + \dots)$$

$$= p q^k \frac{1}{(1-q)}$$

$$= p q^k (1-q)^{-1}$$

$$= p q^k p^{-1}$$

$$\boxed{P(x \geq k) = q^k}$$



And,

$$P(Y \geq t / X \geq k) = \frac{P(Y \geq t \cap X \geq k)}{P(X \geq k)}$$

$$= \frac{P(X - k \geq t \cap X \geq k)}{P(X \geq k)}$$

$$= \frac{P(X \geq k + t)}{P(X \geq k)}$$

$$= \frac{q^{k+t}}{q^k}$$

$$= q^t$$

$$\begin{aligned} \therefore P(Y = t / X \geq k) &= P(Y \geq t / X \geq k) - P(Y \geq t+1 / X \geq k) \\ &= q^t - q^{t+1} \\ &= q^t (1 - q) \\ &= q^t p \end{aligned}$$

$$\therefore P(X = t) = q^t p$$

Change of origin

Since the distribution of $Y = t$ given $X \geq k$ is independent of k . i.e.

i.e. The waiting time k is forgotten. This shows that the change of origin does not affect the geometric distribution. This property of GD is called lack of memory or memoryless property of geometric distribution.

Example: Fit a negative Binomial distribution to the following data:

no. of emails per day: 0 1 2 3 4

no. of days : 237 99 25 3 1

Also, calculate the expected frequencies.

Solution:

x	f	fx	fx ²
0	237		
1	99		
2	25		
3	3		
4	1		

$$N = 365$$

$$\mu_1' = \bar{x} = \frac{\sum fx}{N} = 0.4438$$

$$\boxed{\mu_2'} = \mu_2' = \frac{\sum fx^2}{N} = 0.6630$$

$$\begin{aligned} \mu_2 = \sigma^2 = V(x) &= \mu_2' - (\mu_1')^2 \\ &= 0.6630 - (0.4438)^2 \\ &= 0.46630 \end{aligned}$$

For NB(K, p)

$$\bar{x} = \frac{Kq}{p} \Rightarrow \frac{Kq}{p} = 0.4438 \quad \text{--- (1)}$$

$$\sigma^2 = \frac{Kq^2}{p^2} = 0.4660 \quad \text{--- (2)}$$

Dividing (1) by (2),

$$p = 0.4438 / 0.4660 = 0.9523$$

$$q = 1 - p = 0.0477$$

Again,

$$\frac{kq}{p} = 0.4438 \Rightarrow k = \frac{p \times 0.4438}{q}$$

$$\Rightarrow \boxed{k=9}$$

$$\text{Now, } p(x=0) = p^k = (0.9523)^9 = 0.6441$$

The other prob. of NB distribution are calculated by using recurrence relation.

$$p(x=x+1) = \binom{x+k}{x+1} q p(x=x)$$

x	$\binom{x+k}{x+1} q$	$p(x=x)$	$E = N \times p(x=x)$
0	$\left(\frac{0+9}{0+1}\right) \times 0.477$ $= 0.4293$	$p(0) = 0.6441$	$= 365 \times 0.6441$ ≈ 235
1	0.2385	$p(1) = 0.4293 p(0)$ $= 0.2765$	101
2	0.1749	$p(2) = 0.2385 \times p(1)$ $= 0.0659$	24
3	0.1431	$p(3) = 0.1749 \times p(2)$ $= 0.0115$	4
4	—	$p(4) = 0.1431 \times p(3)$ $= 0.0016$	1 $E = 365$

Example: In a certain manufacturing process, it is known that 1% items on average is defective.

① Find the prob. that 5 items are examined before an item is found to be defective.

② Find the prob. that less than 3 non-defective item before the first defective item.

Solⁿ:

we have,

$$P(X=x) = q^x p$$

① $x+1=5$

$x=4$

$p=0.01$, $q=0.99$

$$P(X=4) = (0.99)^4 (0.01)$$

$$\begin{aligned} \textcircled{12} \quad P(X < 3) &= \sum_{x=0}^2 P(X=x) = P(X=0) + P(X=1) + P(X=2) \\ &= (0.99)^0 (0.01) + (0.99)^1 (0.01) + (0.99)^2 (0.01) \\ P(X < 3) &= 0.297 \end{aligned}$$